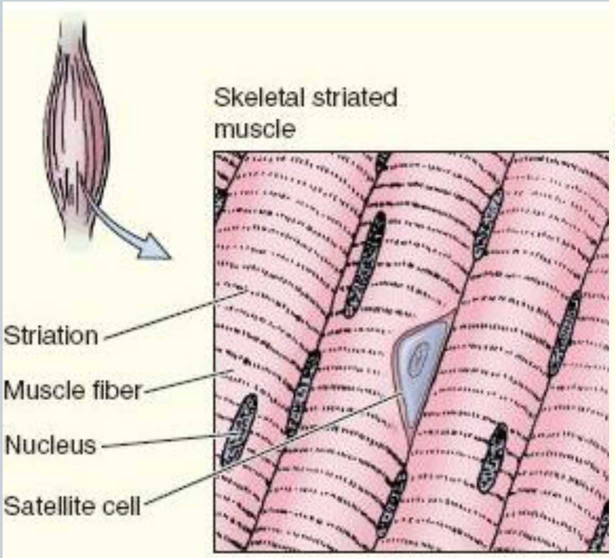
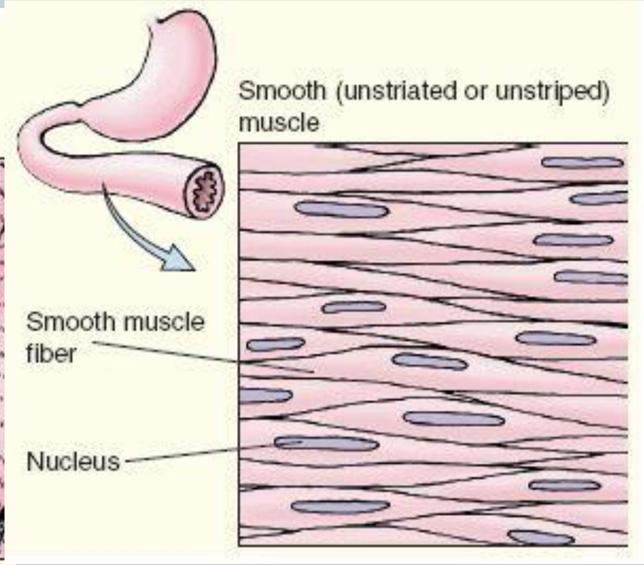
Introduction to muscular system, main muscles of body.

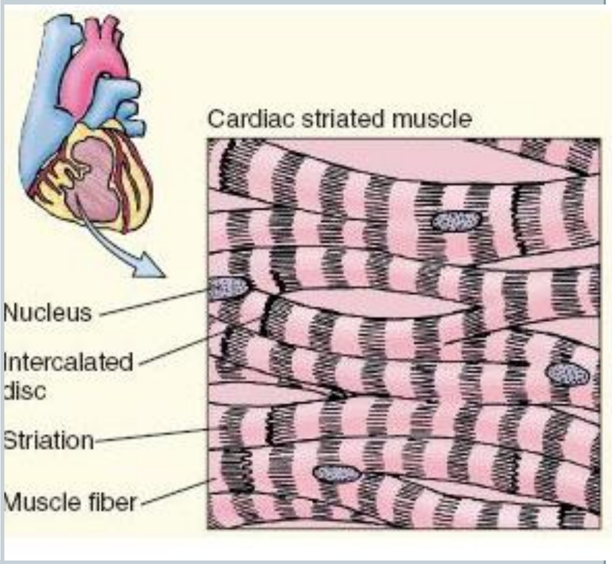
HMU, COLLEGE OF MEDICINE
HDSF 2025\2026

Muscles

- There are 3 types: skeletal, smooth, and cardiac muscles
- they are contractile.







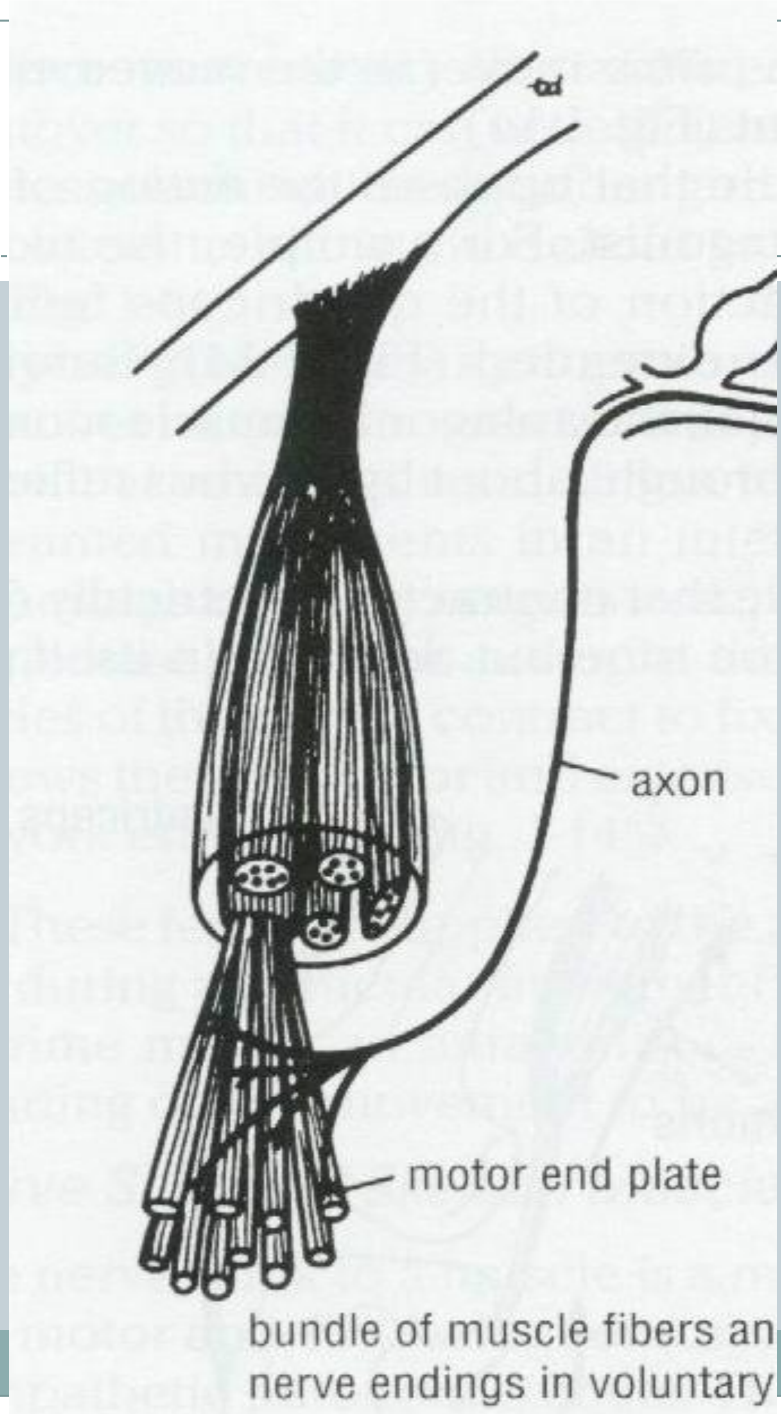
Skeletal muscles

- They produce the movements of the skeleton, they are voluntary and striated. The structural unit is the muscle fibers or cells….the functions:
- 1. produce movement
- 2. maintains posture
- 3. stabilizes joints
- 4. generates heat

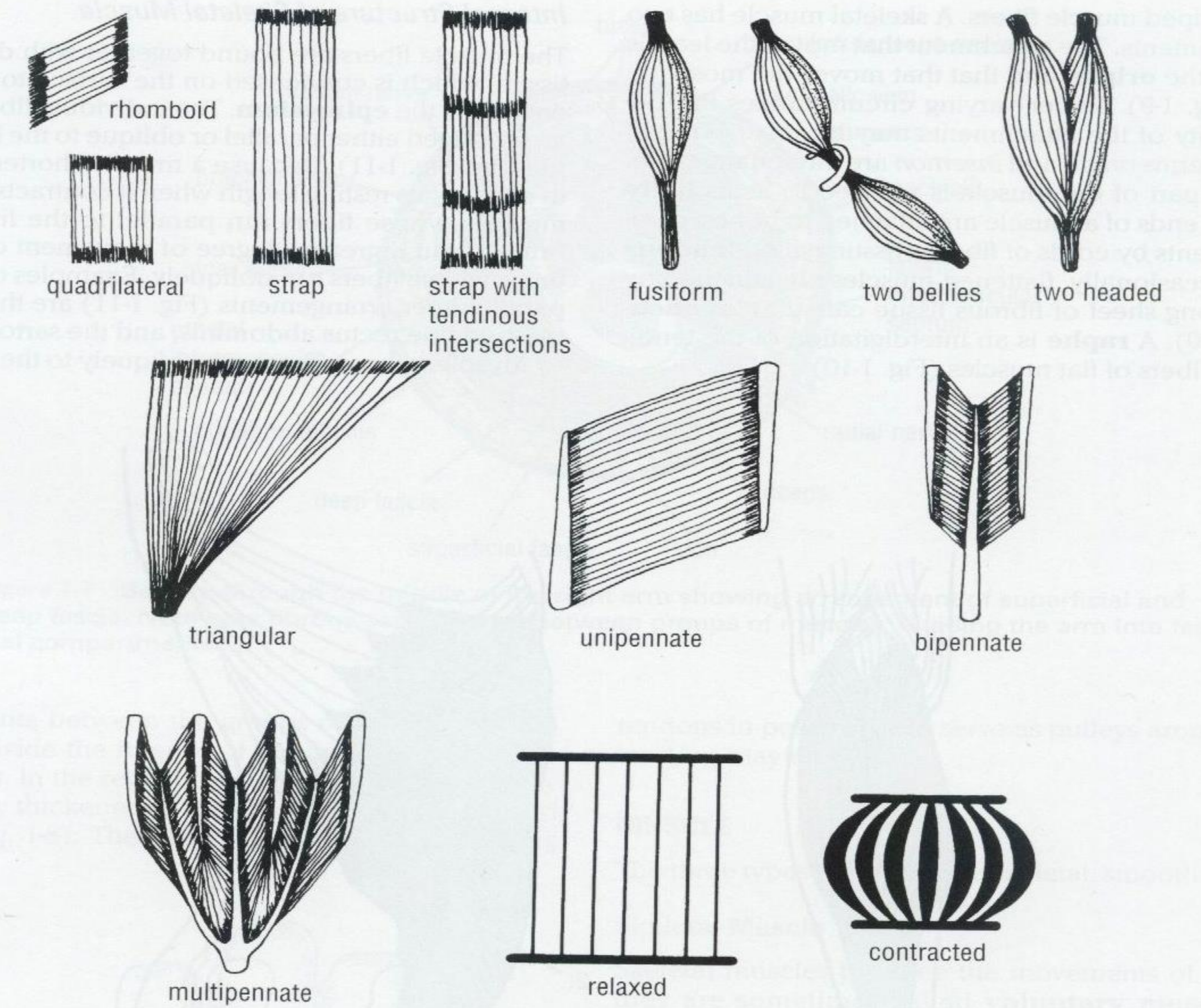
- Skeletal muscles have two or more attachments
- The attachment that moves the least is called origin
- The attachment that moves the most is called the insertion.
- The fleshy part is called muscle belly
- The ends of the muscle which is attached to the skeleton are called tendons.
- Occasionally flattened muscles are attached by a thin sheet of fibrous tissue called aponeurosis.

Internal structure of skeletal muscles

- The structural unit is muscle fiber.
- Each muscle fiber is surrounded by connective tissue layer called endomysium.
- Groups of of muscle fibers (fasciculi) have a connective tissue sheath called perimysium.
- The outer surface of a skeletal muscle have a connective tissue sheath called epimysium.


axon
motor end plate
bundle of muscle fibers an
nerve endings in voluntary

- The muscle fibers are parallel to the long axis of the muscle as in sternocledomastoid
- The fibers may be oblique to the line of pull in pennate muscles
- Pennate muscles are either unipennate, bipennate or multipennate.
- Pennate muscles are stronger but with less range of movement for a given volume of muscle.



ure 1-11 Different forms of internal structure of skeletal muscle. A relaxed and a contracted scle are also shown; note how the muscle fibers, on contraction, shorten by one third to one f of their resting length. Note also how the muscle swells.

Naming of skeletal muscles:

- According to shape:

Deltoid , delta shaped

Trapezius, trapezium in shape, rhomboid, rhombic

2. size, major, minor

3. Number of heads, triceps..biceps

4. action: extensor, flexor

5. attachment: sternocledomastoid

6. depth: superficialis,,profundus

7. length: longus …brevis

Nerve supply of skeletal muscles

- Nerves supplying the skeletal muscles are mixed nerves.
- Motor fibers supply the muscle fibers and stimulate them to contract.
- Each motor nerve fiber will divide into a number of branches and supply several muscle fibers.
- The site where a nerve fiber meets a muscle fiber to supply it is called motor end plate.
- Longer muscle fibers may have more than one motor end plate.

Nerve supply of skeletal muscles

- A motor unit consists of a motor neuron and all muscle fibers that it innervates.
- The sensory fibers arise from specialized sensory endings lying within the muscle or tendons called muscle spindles and tendon spindles respectively.
- These are stimulated by tension in the muscle which may occur during active contraction or passive stretching.

Nerve supply of skeletal muscles

- The nerve supplying the muscle also contains sympathetic fibers for the wall of its blood vessels for regulation of blood flow to the muscle.
- If a nerve trunk of a muscle is severed the muscle can not contract and lead to paralysis of the muscle and there will be loss of the muscle tone in that muscle.

Muscle tone

- While resting every skeletal muscle is in a partial state of contraction, this condition is referred to as muscle tone.

Smooth muscle

- They are involuntary muscles.
- They are non-striated when examined by microscope.
- In the tubes of the body it provides the motive power for propelling the contents.
- In the wall of blood vessels control the caliber of these vessels.
- They are made to contract by nerve impulses from the autonomic nervous system or by hormonal stimulation, or by local stretching of the fibers.

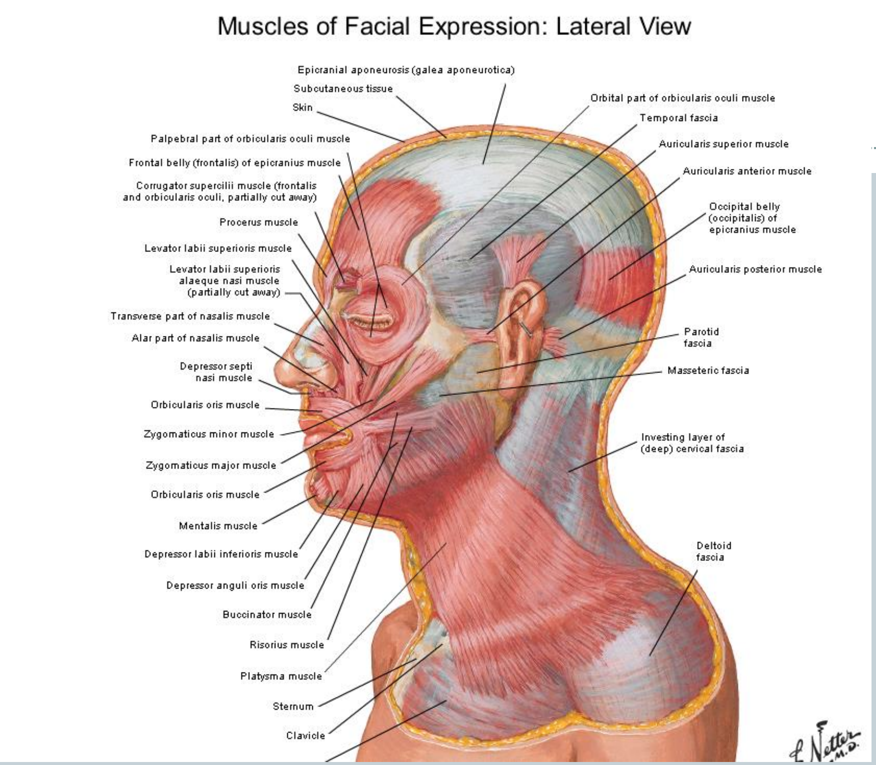
Cardiac muscle

- They are involuntary.
- They are striated.
- Present in the myocardium of the heart.
- They have the property of spontaneous and rhythmic contraction.

Muscles of facial expression

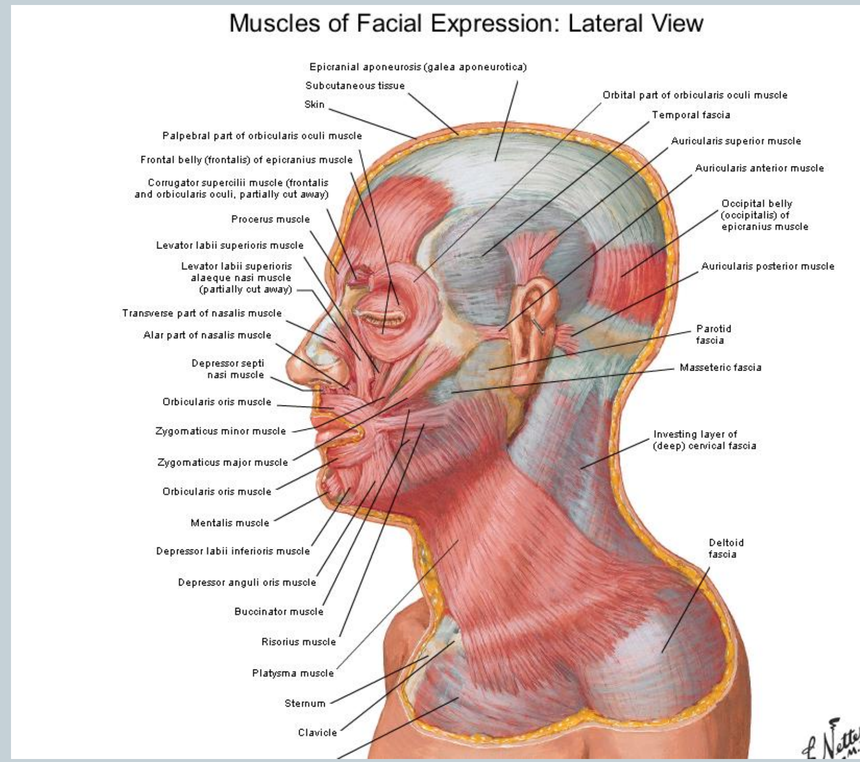
- They provide us the ability to express emotions.
- They lie within the superficial fascia of the face, originate in fascia or bones of skull and insert into the skin.
- They move the skin rather than joints.
- They may act as sphincters (those which encircle the orifices of the face such as eyes , nose , mouth)
- They may act as dilators, opening the orifices.

- The orbicularis oculi is a sphincter of the eye lids,
- levator palpebrae superioris opens the eye lid.

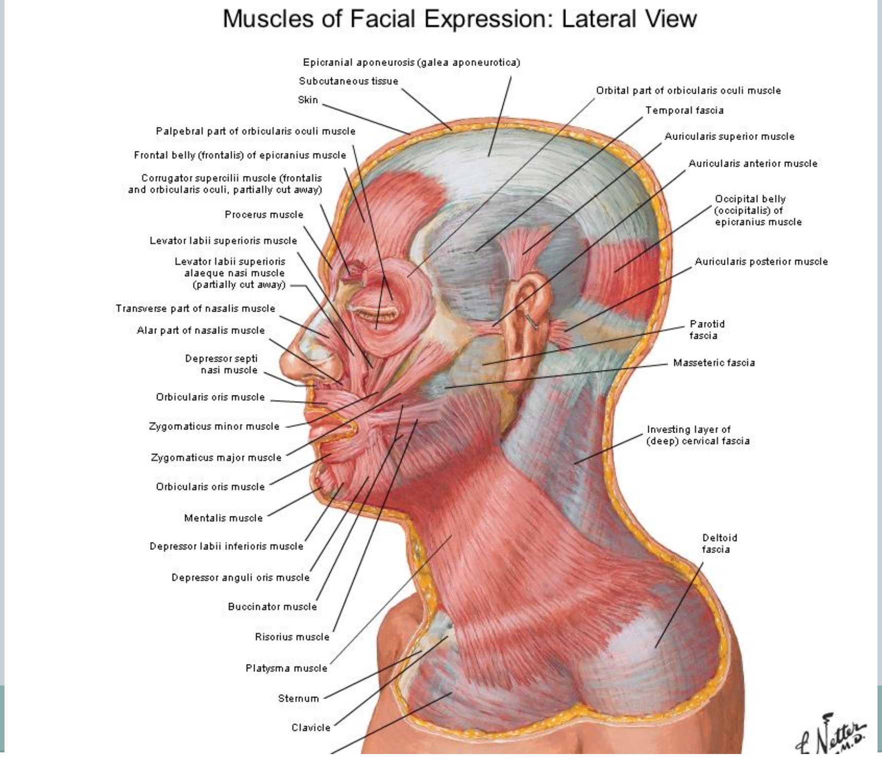


Muscles of the cheeks and lips

- The sphincter muscle is orbicularis oris.
- The dilator muscles are plenty.
- Buccinator muscle forms the major muscular portion of the cheek, it compresses the cheek during blowing.



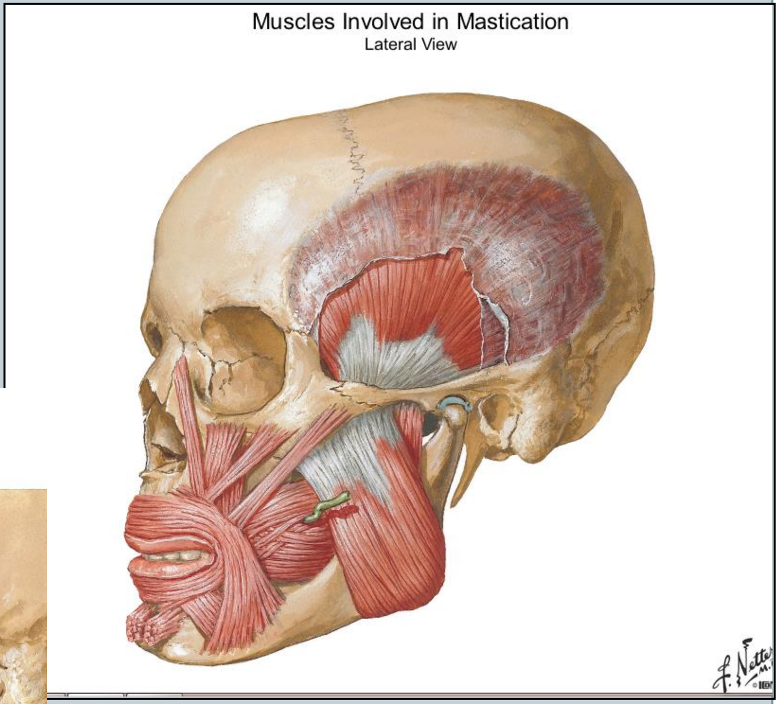
- Occipitofrontalis is made up of two parts, it is the muscle of the scalp.

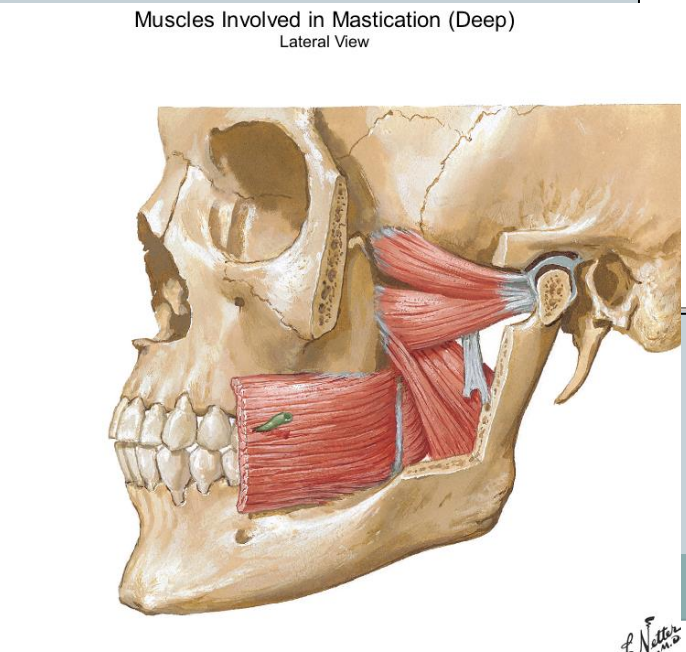


- All the muscles of facial expression are supplied by the facial nerve which is the 7^{th} cranial nerve.

Muscles of mastication

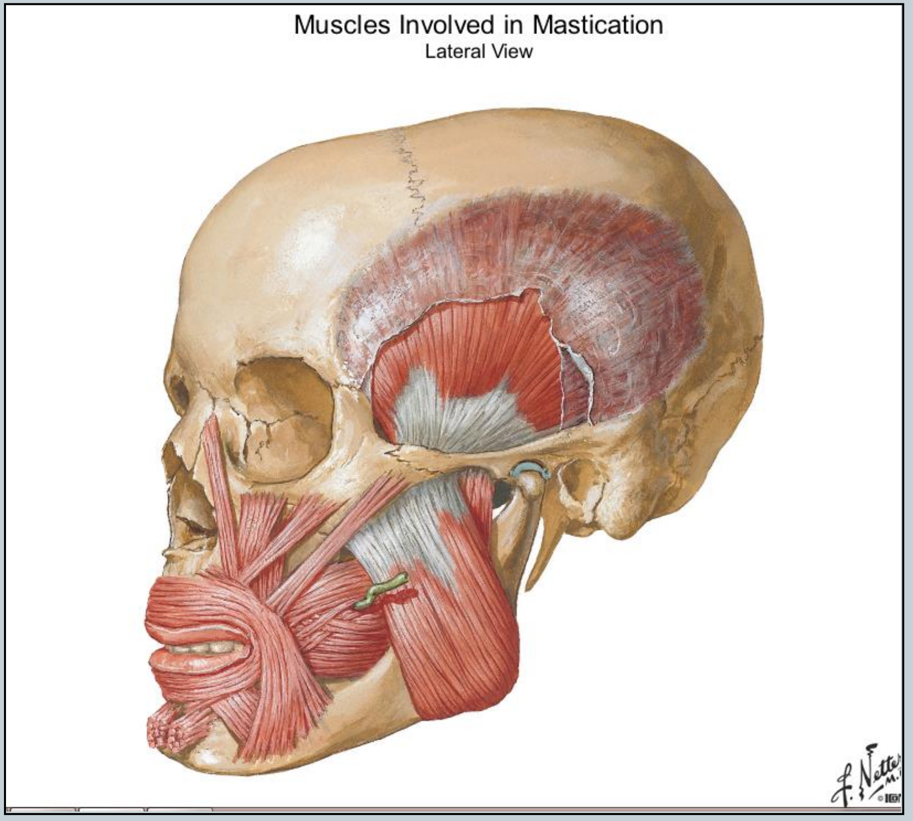
- They move lower jaw bone (chewing)
- They are four pairs: masseter, temporalis, medial pterygoid, lateral pterygoid.





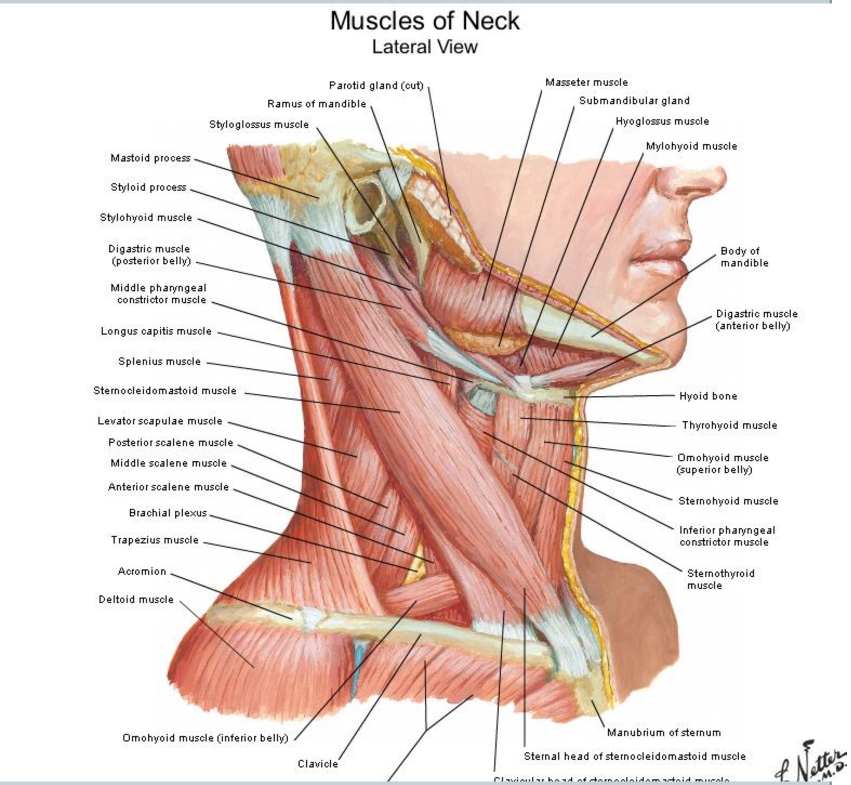
Muscles of mastication

- All are supplied by mandibular division of trigeminal nerve.



Sternocleidomastoid

- It arise from clavicle and sternum and is inserted into mastoid process of temporal bone.
- Its motor supply is by accessory nerve (11^{th} CN)
- It is an important landmark in the neck, that divides the neck into anterior and posterior triangles.



Muscles of Upper Limb

- **1. Muscles lying on anterior & lateral chest wall:**

- **Pectoralis Major:**
- **Pectoralis minor:**
- **Serratus Anterior:**

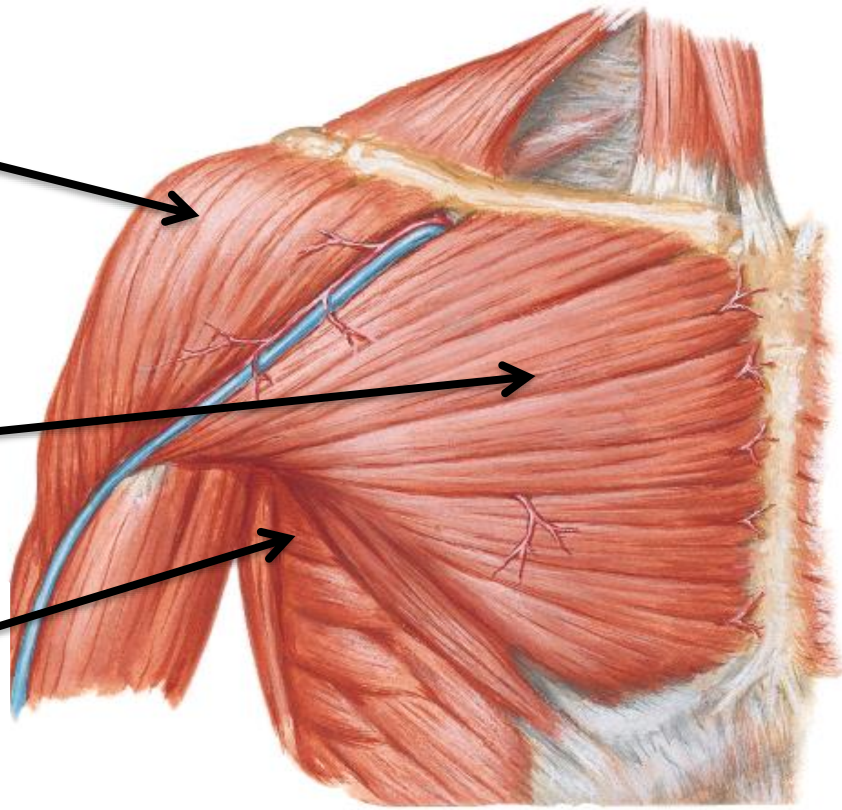
- **2: Muscles lying on the back of trunk:**

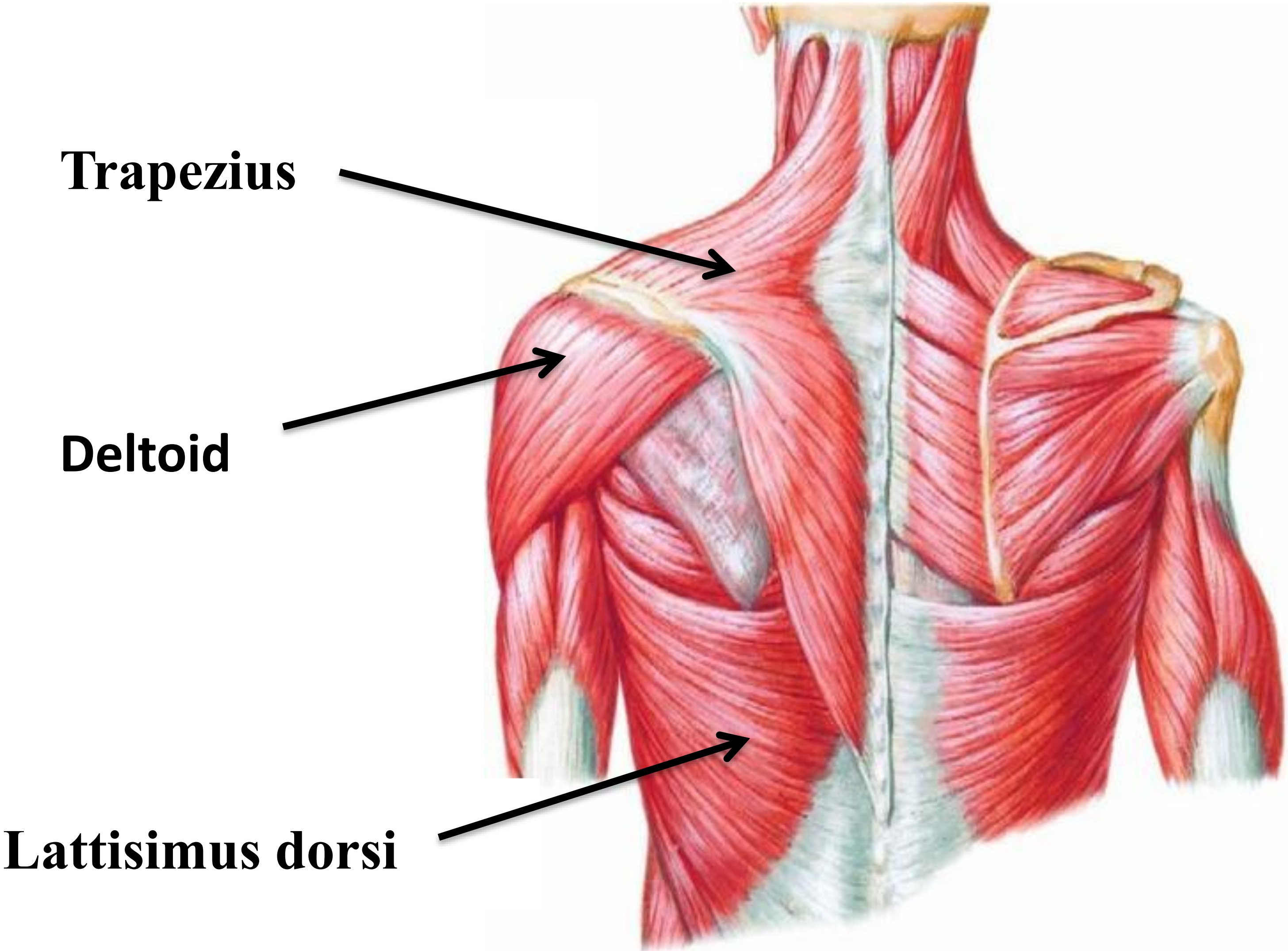
- **Lattisimus dorsi**
- **Trapezius:**

- **3. Muscles around Scapula & Shoulder:**

- **Deltoid:**
- **Supraspinatous:**
- **Infraspinatous:**
- **Teres minor**
- **Teres major:**
- **Subscapularis:**

Muscles of Upper Limb

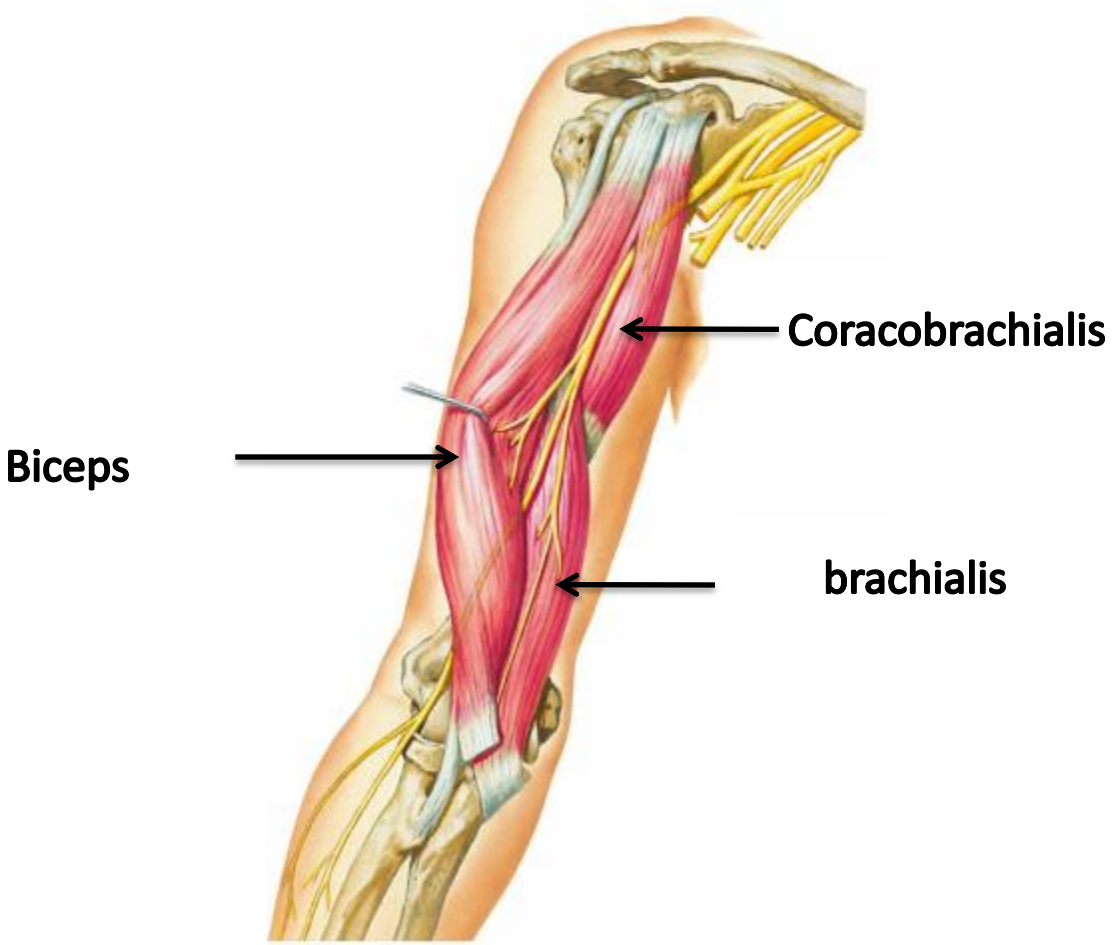


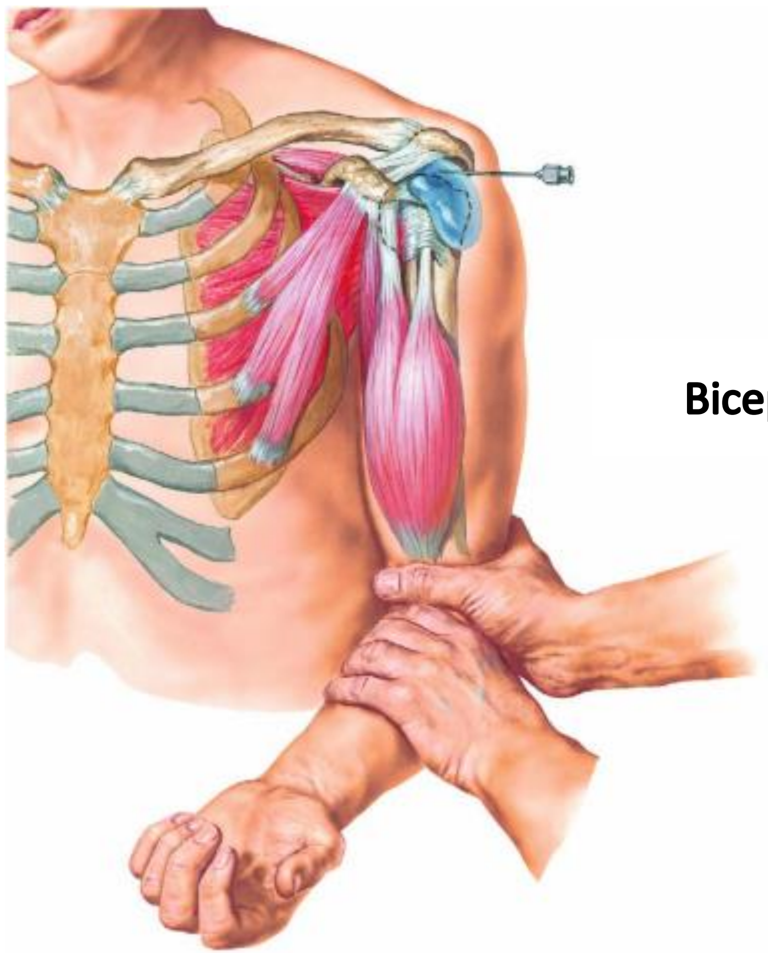

Trapezius
Deltoid
Lattisimus dorsi

Muscles of Arm

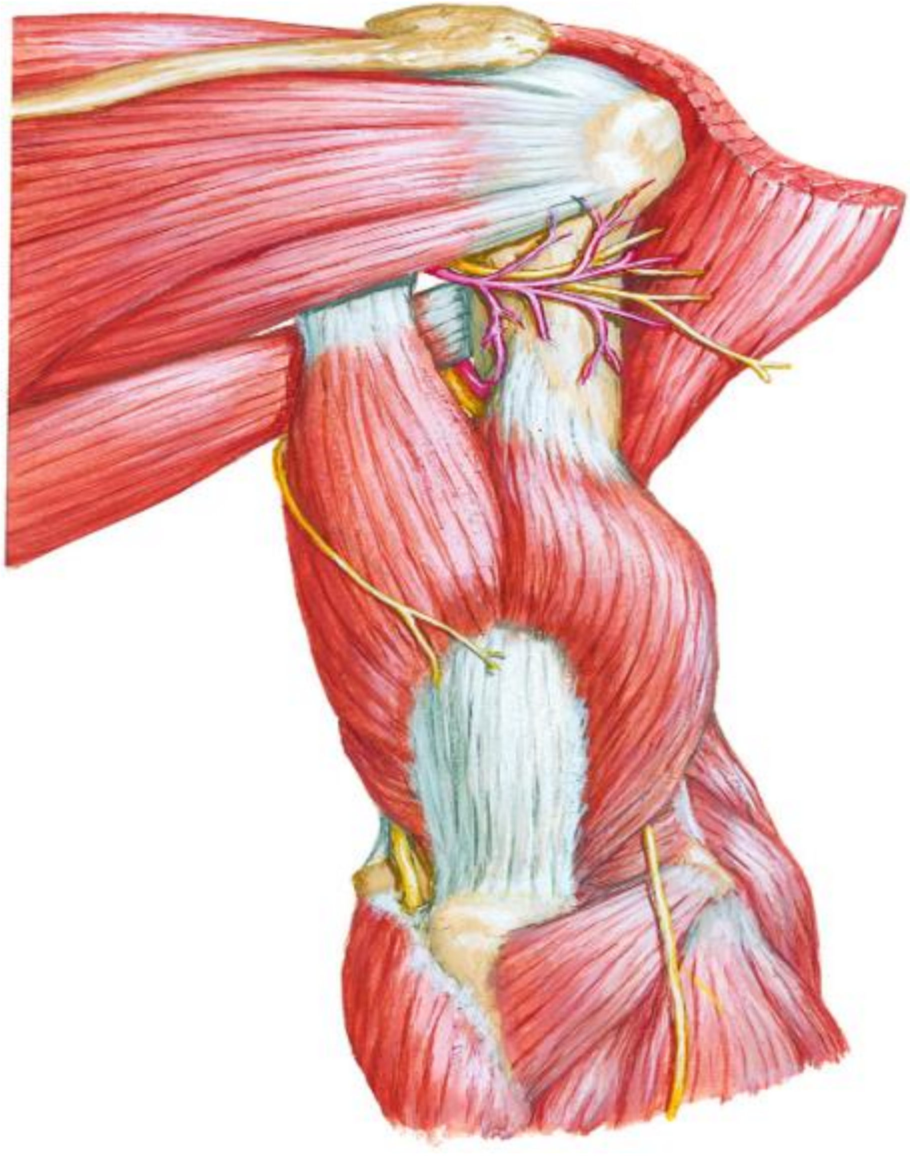
1. Muscles Anterior Compartment of Arm	2. Muscles of Posterior Compartment of Arm
• Coracobrachialis: • Biceps Brachii: • Brachialis:	• Triceps

Muscles Anterior Compartment of Arm





Biceps

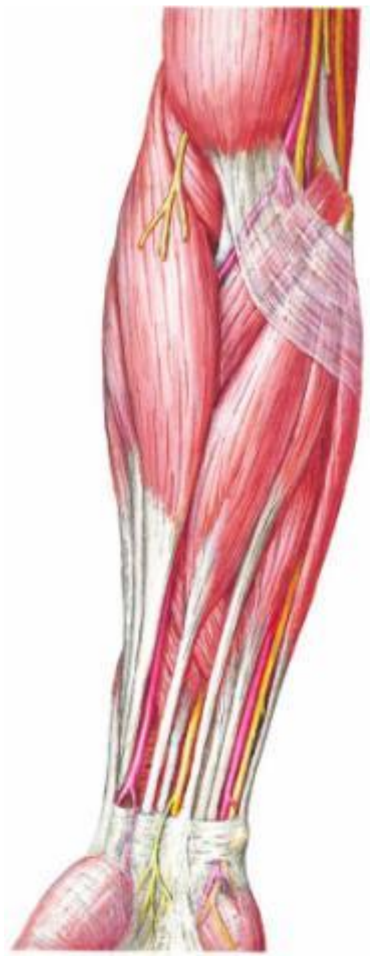


Triceps

Muscles of Forearm

Muscles of Anterior or Flexor Compartment of Forearm	Muscles of Posterior or Extensor Compartment of Forearm:
• ♣ flex the elbow, wrist and interphalangeal joints • ♣ supplied mainly by median nerve and partly from ulnar nerve	• ♣ extends the elbow, wrist and interphalangeal joints • ♣ supplied by radial nerve

Muscles of Anterior or Flexor Compartment of Forearm



Muscles of Posterior or Extensor Compartment of Forearm

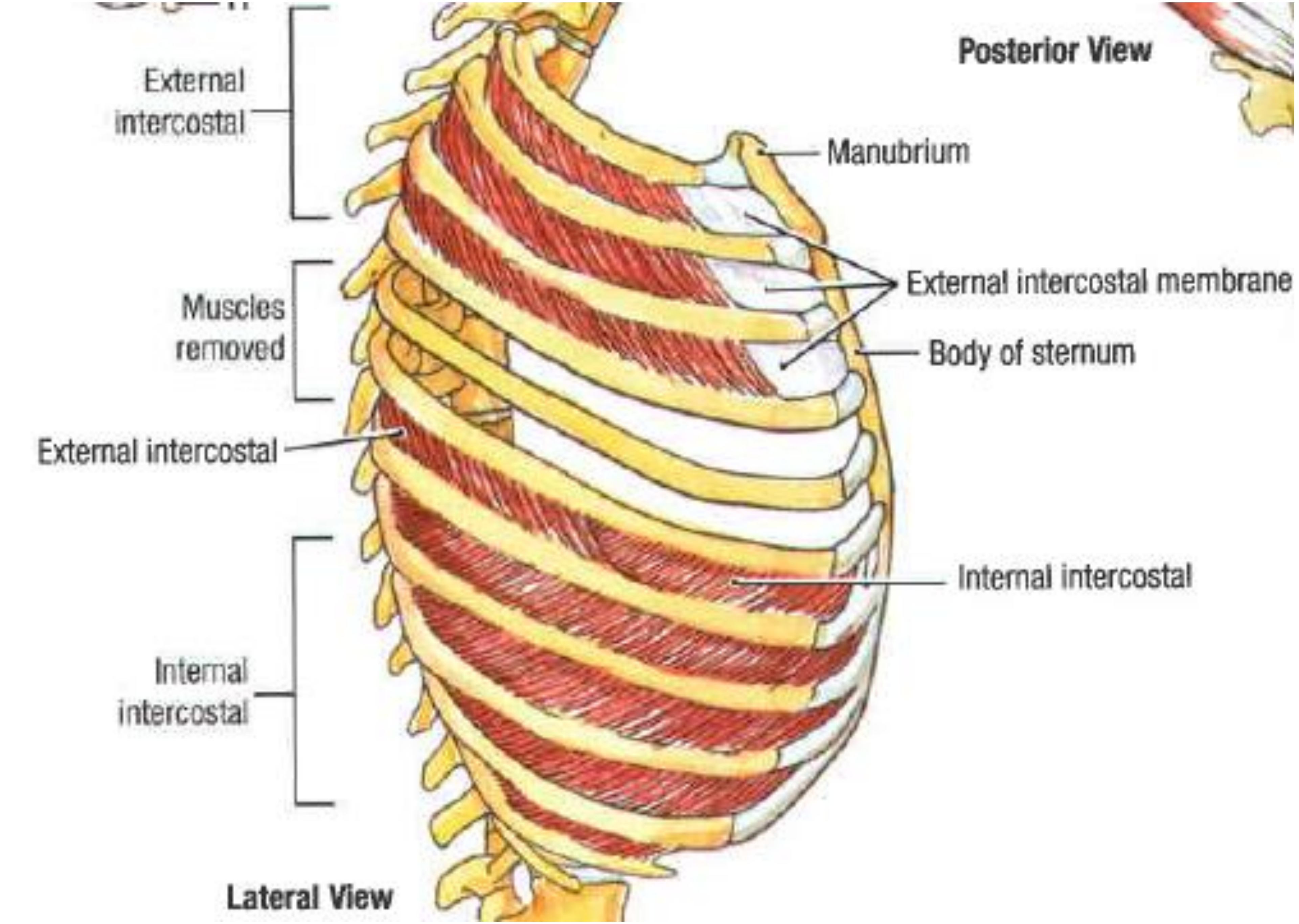
Muscles of Hand

- ♣ 20 in number, small and important for skill movements of hand and fingers
- ♣ supplied by median & ulnar nerves

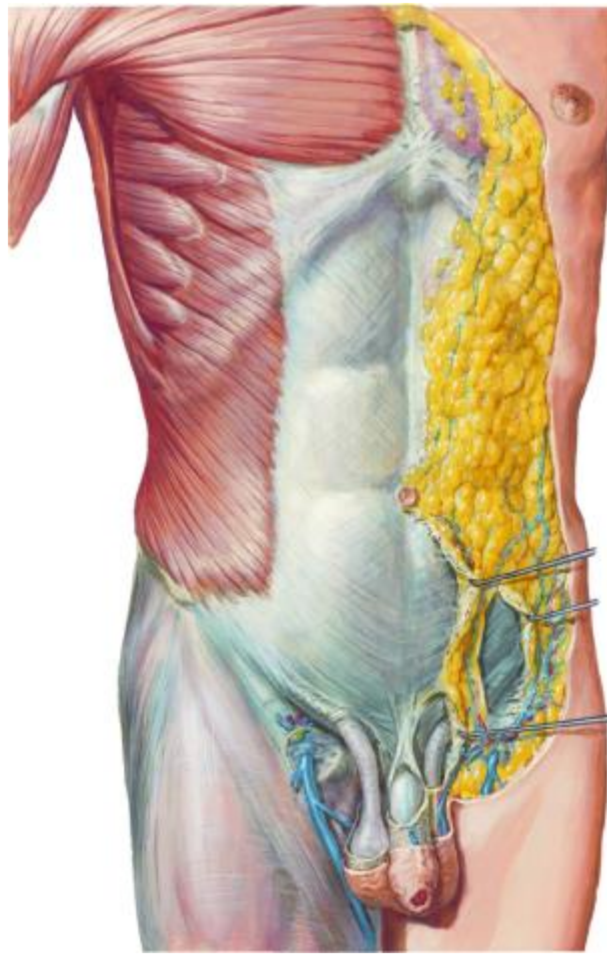
Muscles of Thoracic Wall Proper:

- 1. External intercostals
- 2. Internal intercostals
- 3. Innermost intercostals

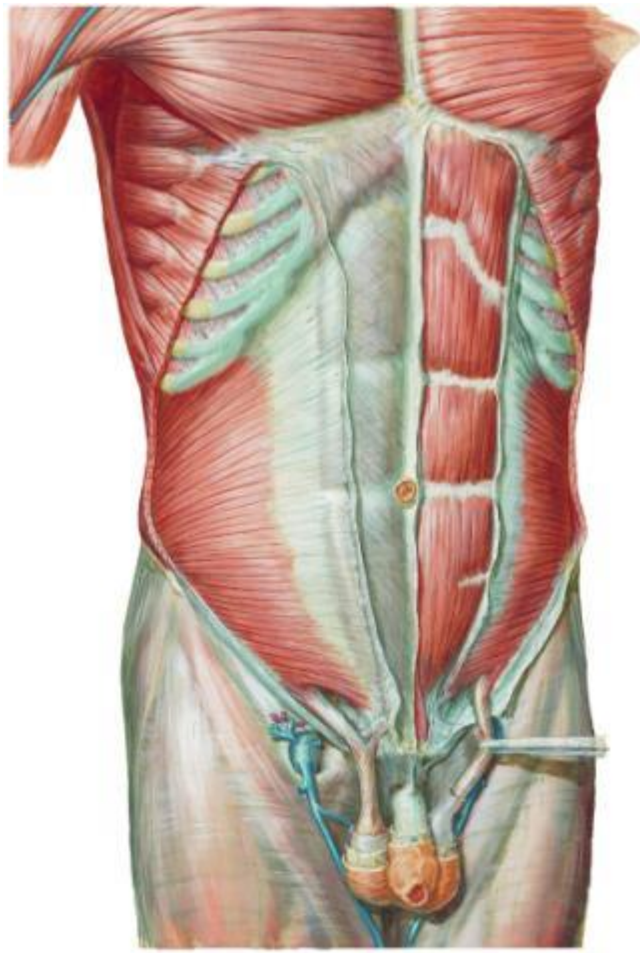
- **Diaphragm**


Posterior View
External intercostal
Manubrium
Muscles removed
External intercostal membrane
Body of sternum
External intercostal
Internal intercostal
Internal intercostal
Lateral View

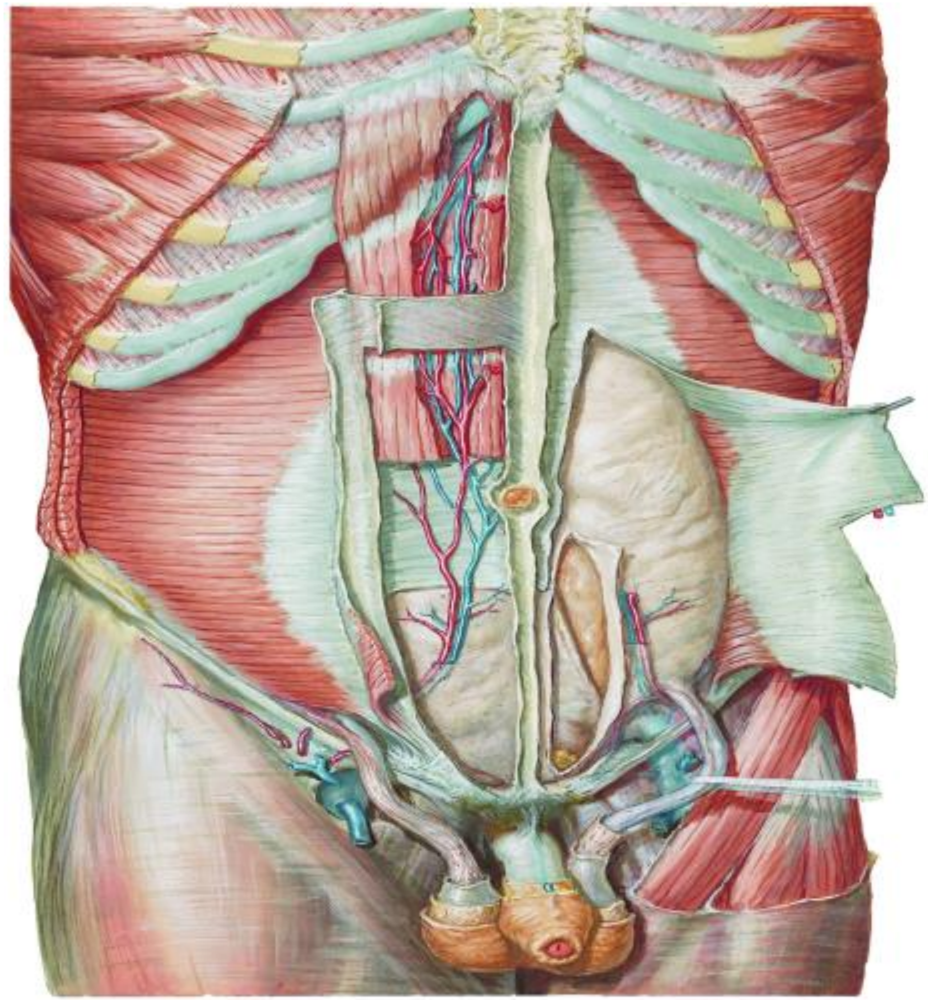
1. External oblique Muscle



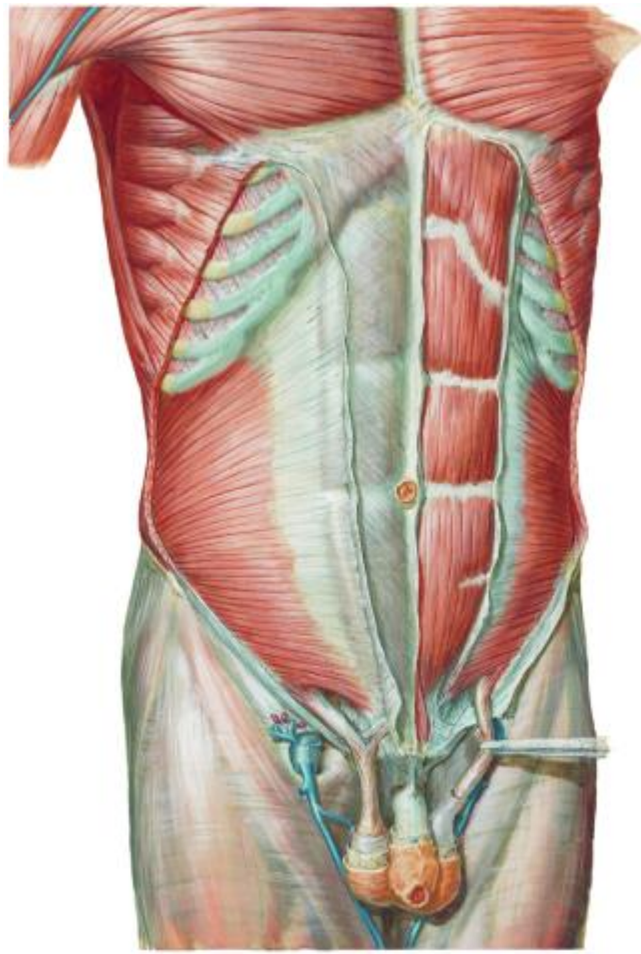
2. Internal oblique Muscle



3. Transverse abdominal Muscle



4- Rectus abdominal Muscle



Muscles of Lower Limb

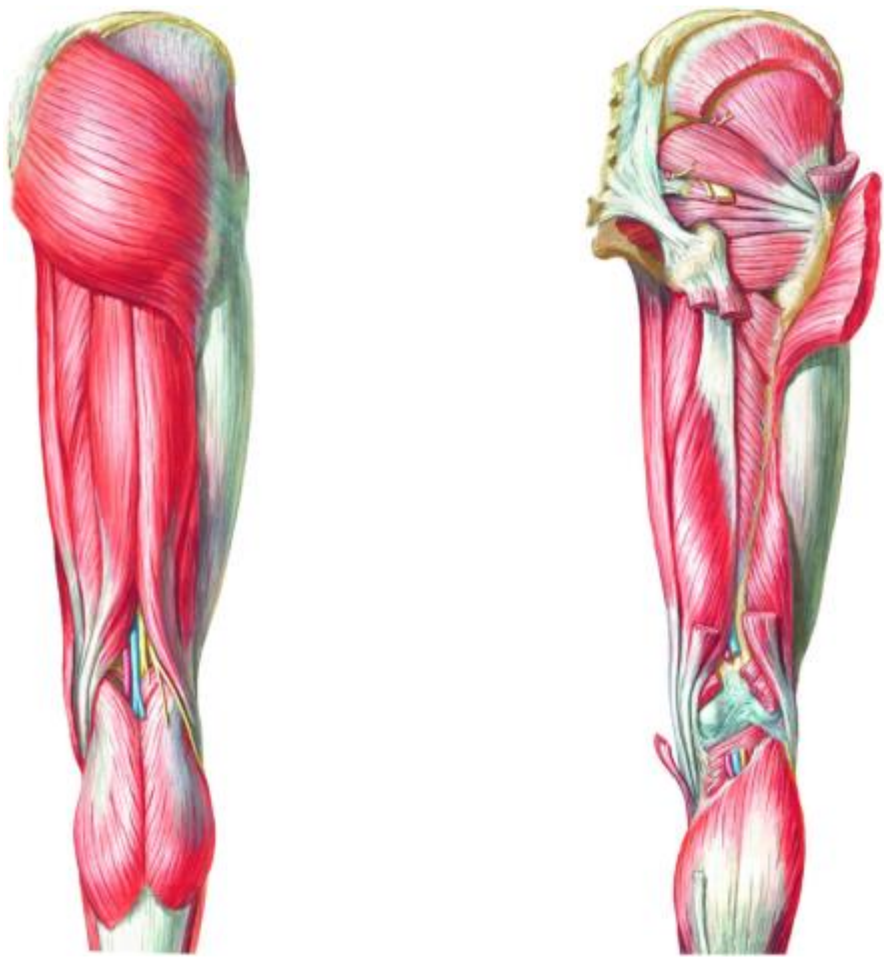
1. Muscles of Gluteal Region

- ♣ include:
- Gluteus maximus,
- gluteus medius &
- gluteus minimus

2- Muscles of Front of Thigh

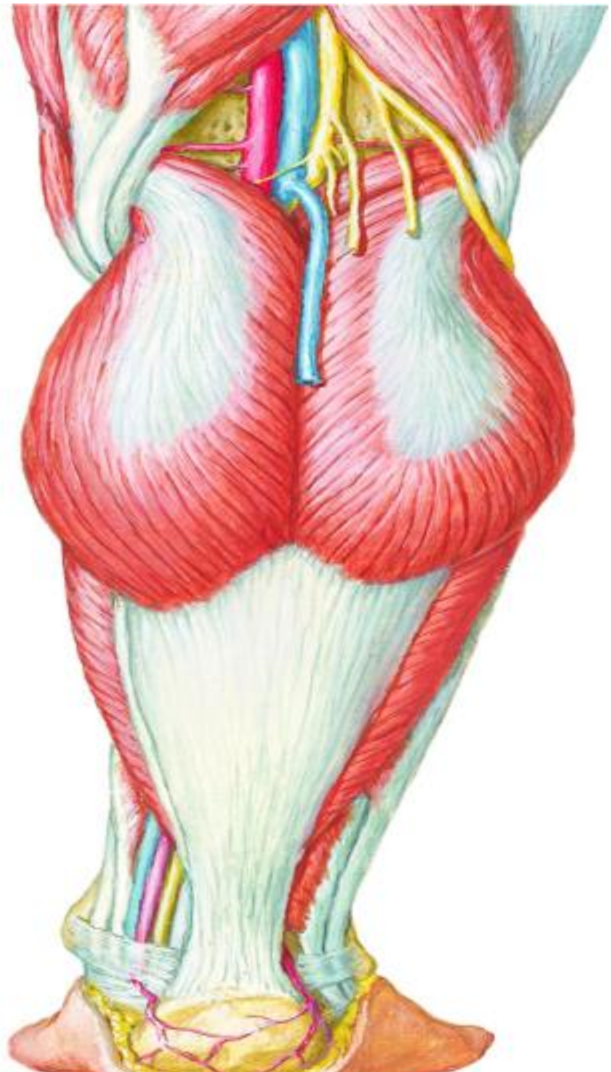
- **Quadriceps femoris muscle:**
- ♣ Consists of:
- **1. rectus femoris**
- **2. vastus medialis**
- **3. vastus intermedius**
- **4. vastus lateralis**
- **Sartorius:**

Gluteus maximus



Quadriceps femoris muscle

Sartorius



Gastrocnemius muscle

Achilles tendon

- QUESTIONS?